

ANSWERS OF MODEL TEST PAPER 4
INTERMEDIATE: GROUP – II
PAPER – 6: FINANCIAL MANAGEMENT & STRATEGIC MANAGEMENT
PAPER 6A : FINANCIAL MANAGEMENT
DIVISION A

1. (c) 18.65%, 16.58%

Ke under two approaches

Calculation of Ke (Using Gordon's Model)

$$K_e = \frac{D_1}{P_0} + g$$

Share Price has grown from 150 to 301 in 5 years,

$$150 (1 + g)^5 = 301.$$

$$(1 + g)^5 = 2.01$$

Therefore, $g = 15\%$, (From Annuity table – Re 1 after 5 years becomes ₹ 2.01 at rate of 15%)

$$D_1 = 8 + 15\% \text{ of } 8 = 9.2$$

P_0 = Average of 52 weeks High price in last 5 years

$$P_0 = (185 + 210 + 252 + 325 + 280) / 5 \\ = 252.40$$

$$K_e = 9.2 / 252.40 + 0.15 \\ = 18.65\%$$

Calculation of Ke (Using CAPM)

$$K_e = R_f + (R_m - R_f) \times \text{Beta} \\ = 8 + (11 \times 0.78) \\ = 16.58\%$$

2. (a) 17.82%

Overall Ke for the company

Approach	Cost of Equity (k)	Weight (w)	K x w
Gordon's	18.65%	0.6	11.19%
CAPM	16.58%	0.4	6.63%
			Total Ke = 17.82%

3. (b) 12%

Intrinsic Value of Debentures today is ₹ 9,740

WN 1 – Calculation of the Pattern of Future Cash flows

YR	PRINCIPAL (I)	INTEREST (II) = Coupon Rate = 9.5% (7.5% + 2%)	PV OF (I + II) @ 10%	PV OF (I + II) @ 15%
1	1,500	997.50	2270.45	2171.74
2	1,500	855	1946.28	1780.72
3	1,500	712.5	1662.28	1454.75
4	1,500	570	1413.84	1183.53
5	1,500	427.50	1196.83	958.31
6	1,500	285	1007.59	771.70
7	1,500	142.50	842.86	617.48
			10340.13	8938.23

$$= 10\% + \frac{(10,340.13 - 9,740)}{(10,340.13 - 8,938.23)} \times 5\% = 12.14\% = 12\% \text{ (approx.)}$$

4. (c) 16.07%, ₹ 87,75,000

$$K_o = W_d \times K_d + W_e \times K_e$$

$$= 0.3 \times 12 + 0.7 \times 17.82$$

$$= 16.07\%$$

Purchase Consideration using M-Cap method

$$= 1,30,000 \text{ eq shares} \times 45 \text{ MPS} \times 1.5X$$

$$= ₹ 87,75,000$$

5. (d) ₹ 66,58,997

It is to be paid equally over 5 years and first instalment is to be paid immediately at Yr 0

Discount rate will be the K_o calculated as above of the company and not 15% which is K_o of Prestige Limited

Year	Amount each year	PV @ 16.07%	PV (₹)
0	17,55,000	1.0000	17,55,000
1	17,55,000	0.8615	15,11,933
2	17,55,000	0.7423	13,02,737
3	17,55,000	0.6395	11,22,323
4	17,55,000	0.5510	9,67,005
	TOTAL PV		66,58,997

6. (d) 19.5%

Financial Leverage (FL) indicates % impact in EPS, if EBIT is affected by 12%

FL = Combined Leverage (CL) / Operating Leverage (OL)

CL = 6.5 (Measure of total risk)

OL = 1 / Margin of Safety

Margin of Safety (MOS) = $\frac{\text{Actual Sales} - \text{B.E Sales}}{\text{Actual Sales}}$

MOS = 20 lakhs – 15 lakhs / 20 lakhs = 0.25

Therefore, OL = 1 / 0.25 = 4

So, FL = 6.5 / 4 = 1.625

So % Change in EPS = 12 x 1.625 = 19.5%

7. (c) 1:2

Item	Cost	Weight	Product
Debt	8%	W	8W
Equity	11%	1 – W	11 – 11W
			WACC = 10

Wd = 1/3 and We = 2/3 Debt Equity Ratio = 1/2

8. (c) ₹ 350 Lakhs

Value of Equity = 30 Lakhs ÷ 15% = ₹ 200 Lakhs

Value of Debt = ₹ 150 Lakhs

Value of Firm = 200 Lakhs + 150 Lakhs = ₹ 350 Lakhs

DIVISION B – Descriptive Questions

- 1. (a)**
 1. Cost of Goods Sold = Sales – Gross Profit
 $= ₹ 7,20,000 - 25\% \times ₹ 7,20,000 = ₹ 5,40,000$
 2. Stock Turnover = $\frac{\text{Cost of Goods Sold}}{\text{Average Stock}} = \frac{₹ 5,40,000}{\text{Average Stock}} = 5 \text{ times.}$
 $\text{Average Stock} = \frac{₹ 5,40,000}{5} = ₹ 1,08,000$
 3. Let Opening Stock be x.
 Closing Stock is ₹ 30,000 more than Opening Stock.
 Closing Stock = (x + 30,000)

$$\text{Average Stock} = \frac{x + x + 30,000}{2} = 1,08,000.$$

$$2x = 2,16,000 - 30,000$$

$$x = \frac{1,86,000}{2} = 93,000 = \text{Opening Stock.}$$

$$\begin{aligned}\text{Closing Stock} &= x + 30,000 \\ &= 93,000 + 30,000 = ₹ 1,23,000\end{aligned}$$

$$4. \quad \text{Liquid Ratio} = \frac{\text{Liquid Assets}}{\text{Current Liabilities}} = \frac{\text{Liquid Assets}}{2,40,000} = 1.25.$$

$$\text{Liquid Assets} = ₹ 3,00,000$$

$$\begin{aligned}5. \quad \text{Current Assets} &= \text{Liquid Assets} + \text{Closing Stock} \\ &= ₹ 3,00,000 + ₹ 1,23,000 = ₹ 4,23,000\end{aligned}$$

(b) Calculation of slab wise Overall Cost of Capital

(i)

Project Cost	Capital Source	Weights (w)	Cost (k)	w x k (%)
Upto 5 Lakhs	Debt	0.3	10	3
	Equity	0.7	12	8.4
			Ko	11.4
Above 5 lakhs upto 10 lakhs	Debt	0.3	12	3.6
	Equity	0.7	13.5	9.45
			Ko	13.05
Above 10 lakhs upto 20 lakhs	Debt	0.3	13	3.9
	Equity	0.7	15	10.5
			Ko	14.4
Above 20 lakhs	Debt	0.3	14	4.2
	Equity	0.7	16	11.2
			Ko	15.4

Cost of Raising funds for Project I

Total Capital	Ko(%)	Total Cost (in ₹)
5,00,000	11.40	57,000
5,00,000	13.05	65,250
5,00,000	14.40	72,000
15,00,000		1,94,250

$$\begin{aligned}\text{Overall COC (\%)} &= \text{Total Cost (in ₹)} / \text{Total Capital} \\ &= 1,94,250 / 15,00,000 * 100 \\ &= 12.95 \%\end{aligned}$$

Cost of Raising funds for Project II

Total Capital	Ko(%)	Total Cost (in ₹)
5,00,000	11.4	57,000
5,00,000	13.05	65,250
10,00,000	14.4	1,44,000
6,00,000	15.4	92,400
26,00,000		3,58,650

$$\text{Overall COC (\%)} = 358650 / 2600000 * 100 = \mathbf{13.79\%}$$

- (ii) If any project is expected to give an after-tax return of 13%, it can be accepted only if the maximum Overall COC (%) of that project equals 13% or less, as at 13%, project would be at break-even i.e. earning 13% from the project and incurring 13% COC.

So, under that scenario, Project I can be taken as its COC is 12.95% whereas Project II can't be taken as its COC is 13.79%.

Maximum Value of the Project that can be taken at 13% is approx. (Using IRR technique Interpolation)

At 15 Lakhs Ko = 12.95%

At 26 Lakhs Ko = 13.79%

By interpolation, maximum value of Project at 13% will be

$$15 \text{ Lakhs} + \{(0.05 \times 11) / 0.84\}$$

$$= \mathbf{15.6548 \text{ lakhs}}$$

(c) Income Statement

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBT}} = \frac{\text{EBT} + \text{Interest}}{\text{EBT}} = \frac{\text{EBT} + 2,000}{\text{EBT}} = \frac{2}{1}$$

$$\text{EBT} + ₹ 2000 = 2 \text{ EBT.}$$

$$\text{EBT} = ₹ 2,000$$

$$\text{EBIT} = \text{EBT} + \text{Interest} = ₹ 2000 + ₹ 2000 = ₹ 4,000.$$

$$\text{Contribution} \quad \frac{\text{Contribution}}{\text{EBIT}} = \frac{\text{Contribution}}{4,000} = \frac{3}{1}$$

Sales	$\frac{\text{Contribution}}{\text{PVR}} = \frac{12,000}{25\%}$	48,000
Less: Variable Cost	Given = 75%	(36,000)
Contribution		12,000
Less: Fixed Cost (Contribution - EBIT = ₹ 12,000 – ₹ 4,000)		(8,000)

EBIT	4,000
Less: Interest	(2,000)
EBT	2,000
Less: Tax at 30%	(600)
EAT	1,400

2. (a)

Particulars	Result
Current liabilities	1,56,000
Total Variable expenses = Purchases & Operating Expenses	$1,56,000 \div 60 \times 360 = 9,36,000$
Variable expenses % of Sales	$9,36,000 \div 12,00,000 \times 100 = 78\%$

Particulars	Present	Proposed
1. Sales	$1 \text{ Lakh} \div 30 \times 360$ $= 12,00,000$	$12 \text{ Lakhs} + 1/3^{\text{rd}}$ $= 16,00,000$
2. Variable Cost at 78%	9,36,000	12,48,000
3. Cash Discount	$12 \text{ Lakh} \times 50\% \times 1\%$ $= 6,000$	$16 \text{ Lakh} \times 80\% \times 2\%$ $= 25,600$
4. Bad debts	$12 \text{ Lakh} \times 1.5\%$ $= 18,000$	$16 \text{ Lakh} \times 2\%$ $= 32,000$
5. Profit before Tax	2,40,000	2,94,400
6. Tax @ 30%	72,000	88,320
7. Profit after Tax	1,68,000	2,06,080
8. Opportunity Cost of Invest. in Debtors	$9,36,000 \times 30/360 \times$ $70\% \times 15\% = 8,190$	$12,48,000 \times 20/360 \times$ $70\% \times 15\% = 7,280$
9. Net Benefit	1,59,810	1,98,800

Advise: Proposed policy should be adopted since the net benefit is increased by $(\text{₹ } 1,98,800 - 1,59,810) = \text{₹ } 38,990$.

(b) (i) As per **Gordon's Model**, Price per share is computed using the formula:

$$P_0 = \frac{E_1(1-b)}{K_e - br}$$

Where,

P_0 = Price per share

E_1 = Earnings per share

Payout ratio = $45/180 = 25\%$

b = Retention ratio; $(1 - b = \text{Pay-out ratio}) = 1 - 0.25 = 0.75$

K_e = Cost of capital

r = IRR

br = Growth rate (g)

Applying the above formula, price per share

$$P_0 = \frac{180(1-0.75)}{0.17-0.75 \times 0.2} = \frac{45}{0.02} = ₹ 2,250$$

(ii) As per Walter's Model, Price per share is computed using the formula:

$$\text{Price (P)} = \frac{D + \frac{r}{K_e}(E-D)}{K_e}$$

Where,

P = Market Price of the share.

E = Earnings per share.

D = Dividend per share.

K_e = Cost of equity/ rate of capitalization/ discount rate.

r = Internal rate of return/ return on investment

Applying the above formula, price per share

$$P = \frac{45 + \frac{0.20}{0.17}(180-45)}{0.17}$$

$$\text{Or, } P = \frac{45 + 158.82}{0.17} = ₹ 1,200 \text{ (approx..)}$$

3. (a) Calculation of Present value of cash inflows (PVCi)

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Savings in cost due to Production Delays	-	3,50,000	3,50,000	3,50,000	3,50,000	3,50,000
Savings in Salaries	-	21,00,000	21,00,000	21,00,000	21,00,000	21,00,000
Reduction in lost sales	-	1,75,000	1,75,000	1,75,000	1,75,000	1,75,000
Gain due to timely billing	-	3,25,000	3,25,000	3,25,000	3,25,000	3,25,000
	-	29,50,000	29,50,000	29,50,000	29,50,000	29,50,000
Less:						
Salary of AI specialists	-	13,00,000	13,00,000	13,00,000	13,00,000	13,00,000
Annual Maint. & Op Cost	-	1,80,000	2,00,000	1,20,000	1,10,000	1,30,000
NPBDT	-	14,70,000	14,50,000	15,30,000	15,40,000	15,20,000
(-) Depreciation	-	9,20,000	5,52,000	3,31,200	1,98,720	1,19,232

NPBT	-	5,50,000	8,98,000	11,98,800	13,41,280	14,00,768
(-) Tax @ 25%	-	1,37,500	2,24,500	2,99,700	3,35,320	3,50,192
NPAT	-	4,12,500	6,73,500	8,99,100	10,05,960	10,50,576
(+) Depreciation	-	9,20,000	5,52,000	3,31,200	1,98,720	1,19,232
(+) Annual Maint. & Op Cost	-	1,80,000	2,00,000	1,20,000	1,10,000	1,30,000
Gross Cash Inflows	-	15,12,500	14,25,500	13,50,300	13,14,680	12,99,808
(-) Annual Maint. & Op Cost actually paid	1,80,000	2,00,000	1,20,000	1,10,000	1,30,000	-
Net Cash Inflows	-1,80,000	13,12,500	13,05,500	12,40,300	11,84,680	12,99,808
(+) Sale Value at the end of life	-	-	-	-	-	1,90,000
	-1,80,000	13,12,500	13,05,500	12,40,300	11,84,680	14,89,808
PV Factor @ 12%	1	0.8929	0.7972	0.7118	0.6355	0.5674
PV of Cash Inflows	-1,80,000	11,71,875	10,40,737	8,82,821	7,52,886	8,45,357
Total PV of Cash Inflows	45,13,675					

Calculation of Present value of cash outflows (PVCO)

As mentioned in the question, 75% of the depreciable value will be paid at the beginning. Depreciable value means purchase price plus the installation cost.

	Year 0	Year 1
Purchase Price & Installation Cost	17,25,000	5,75,000
PV Factor @ 12%	1	0.8929
PVCO	17,25,000	5,13,418

(2) Total PVCO = 22,38,418

(3) PV of Tax on Capital Gains (Only asset in the block) - 5th Year end

Capital Gains = Sale Price (-) Closing WDV at 5th year

= 1,90,000 (-) 1,78,848

= 11,152

Tax @ 20% on above = 2230.40

PV = 2,230.40 x 0.5674 = 1,266

Net PVCI = PVCI - PV of Tax on Capital Gains

= 45,13,675 - 1,266 = 45,12,409

NPV = Net PVCI – PVCO

= 45,12,409 - 22,38,418

$$= 22,73,991$$

$$(II) \quad PI = PVCI / PVCO = 45,12,409 / 22,38,418 = 2.0158$$

$$(III) \quad ARR = \text{Average NPAT} / \text{Initial Investment} \\ = 8,08,327.2 / 23,00,000 \times 100 = 35.145\%$$

Note – ARR is calculated based on Initial Investment, similarly it can be calculated based on Average Investment

(b) Lintner's model has two parameters:

- i. The target payout ratio,
- ii. The spread at which current dividends adjust to the target.

4. (a) Normally it is considered that the trade credit does not carry any cost. However, it carries the following costs:

- (i) **Price:** There is often a discount on the price that the firm undergoes when it uses trade credit, since it can take advantage of the discount only if it pays immediately. This discount can translate into a high implicit cost.
- (ii) **Loss of goodwill:** If the credit is overstepped, suppliers may discriminate against delinquent customers if supplies become short. As with the effect of any loss of goodwill, it depends very much on the relative market strengths of the parties involved.
- (iii) **Cost of managing:** Management of creditors involves administrative and accounting costs that would otherwise be incurred.
- (iv) **Conditions:** Sometimes most of the suppliers insist that for availing the credit facility the order should be of some minimum size or even on regular basis.

- (b) (i) **Fully Hedged Bonds:** In foreign bonds, the risk of currency fluctuations exists. Fully hedged bonds eliminate the risk by selling in forward markets the entire stream of principal and interest payments.
- (ii) **Medium Term Notes (MTN):** Certain issuers need frequent financing through the Bond route including that of the Euro bond. However, it may be costly and ineffective to go in for frequent issues. Instead, investors can follow the MTN programme. Under this programme, several lots of bonds can be issued, all having different features e.g. different coupon rates, different currencies etc. The timing of each lot can be decided keeping in mind the future market opportunities. The entire documentation and various regulatory approvals can be taken at one point of time.
- (iii) **Floating Rate Notes (FRN):** These are issued up to seven years maturity. Interest rates are adjusted to reflect the prevailing exchange rates. They provide cheaper money than foreign loans.

- (iv) **Euro Commercial Papers (ECP):** ECPs are short term money market instruments. They have maturity period of less than one year. They are usually designated in US Dollars.
- (c) DOL can never be between zero and one. It can be zero or less or it can be one or more.

When Sales is much higher than BEP sales, DOL will be slightly more than one. With decrease in sales, DOL will increase. At BEP, DOL will be infinite. When sales is slightly less than BEP, DOL will be negative infinite. With further reduction in sale, DOL will move towards zero. At zero sales, DOL will also be zero.

OR

The finance executive of an organisation plays an important role in the company's goals, policies, and financial success. His responsibilities include:

- (a) **Financial analysis and planning:** Determining the proper amount of funds to employ in the firm, i.e. designating the size of the firm and its rate of growth.
- (b) **Investment decisions:** The efficient allocation of funds to specific assets.
- (c) **Financing and capital structure decisions:** Raising funds on favourable terms as possible i.e. determining the composition of liabilities.
- (d) **Management of financial resources** (such as working capital).
- (e) **Risk management:** Protecting assets.